

CANDOR: SCORE FOLLOWING AS SEQUENTIAL SELECTION OVER NOTE HYPOTHESES

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ABSTRACT

We formulate real-time score following on sheet-music images as sequential selection: at each instant the page affords a set of note hypotheses, and following the performance means choosing among them under a temporal constraint, rather than regressing a coordinate. CANDOR realizes this with three parts, a candidate generator, a parameter-free transition prior over displacements, and a 23k-parameter permutation-invariant scorer trained listwise on synthesized audio. On real room-microphone piano recordings CANDOR tracks 94.4% of onsets within 0.5 s and runs in 1.25 ms per step on one CPU thread. The formulation explains where accuracy on real audio is lost. A candidate oracle shows that a correct hypothesis is always present and a causal selector could reach 99.6%, so the synthetic-to-real gap is a decision problem, not a perception problem; consistent with this, three different image representations for the scorer, including a self-supervised vision transformer and none at all, give identical accuracy, whereas removing the temporal history costs 7.2 points. Because selection is separable from proposal, one trained selector improves every candidate generator we attach it to, by 20.3, 11.0 and 7.9 points. We further report that accuracy on synthesized validation audio is nearly uninformative about accuracy on real audio ($r = 0.16$ over 48 checkpoints), and that the forward-motion prior which makes selection work is also what makes score discontinuities hard.

Index Terms— Score following, sheet music, real-time alignment, set prediction, music information retrieval

1. INTRODUCTION

Score following, the real-time alignment of a live performance to its score, underlies automatic page turning [1] and computer accompaniment [2, 3]. Most music exists only as printed or scanned sheets, and optical music recognition remains error-prone [4], so a line of work follows the performance directly on the sheet image [5, 6, 7, 8, 9, 10, 11]. These systems are trained on audio synthesized from the score and lose a large part of their accuracy on real recordings, which is the field’s main open problem.

Almost all of this work treats the task as *regression*: a network maps audio and a page to a coordinate, whether as a

heatmap, a bounding box or a reinforcement-learning action. We argue that the natural formulation is *selection*. A page of music affords a finite, discrete set of places the performance could be at, one per notehead. Following the score is choosing among them, again at every instant, subject to the constraint that the choices form a plausible trajectory. Regression asks a network to synthesize a number that a set already contains.

The distinction is not cosmetic, and one measurement makes it concrete. Given a set of 256 note hypotheses per instant, we find that on real room-microphone piano recordings a hypothesis within the 0.5 s tolerance is present at every one of 4,149 scored onsets, and that an oracle which must commit one step at a time and may never move backwards tracks 99.6% of them. A tracker built on the same hypotheses that reports the most confident one tracks 80.0% (Fig. 1). Nineteen of the twenty points lost to real audio are lost in the decision, not in the perception. Under a regression view this is invisible, because there is no set to take an oracle over.

CANDOR is a score follower built on this formulation. Its three parts are a candidate generator, a transition prior with no trained parameters, and a small permutation-invariant scorer that judges each hypothesis against the others in its instant. Because selection is separable from proposal, the generator is an interchangeable component: we attach the same trained selector to three different published note detectors and it improves each of them, which is a property of the formulation rather than a tuning result.

Our contributions are: (i) a formulation of sheet-image score following as sequential selection over note hypotheses, with a candidate-oracle analysis that locates the synthetic-to-real gap in the decision; (ii) CANDOR, a tracker realizing it, which reaches 94.4% of onsets within 0.5 s on real recordings at 1.25 ms per step and transfers unchanged across candidate generators; (iii) three findings the formulation exposes and regression hides: the image representation the selector reads is irrelevant while its temporal history is essential, synthetic validation accuracy barely ranks models for real audio, and the forward-motion prior that makes selection work is what makes discontinuities hard.

Related work. Symbolic score followers already select, over score states, with hidden (semi-)Markov models [12, 13] or online time warping [14]; our formulation brings that stance to the sheet image, where the state set must be pro-

■ detector argmax (2.3 s off) ground truth
■ ours (0.05 s off) — top-40 candidates



Fig. 1. One instant of a real room-microphone recording. Grey: 40 of the note hypotheses the page affords. Reporting the most confident (red) lands on the wrong staff; CANDOR (teal) selects a less confident hypothesis on the true note (ring). The right answer was in the set and was not chosen.

posed from pixels rather than read from a symbolic score. Sheet-image trackers regress: staff strips [5], reinforcement-learning agents [6, 7], full-page segmentation [8], and audio-conditioned detection [9, 10] with FiLM [15], whose box proposals we reuse as a candidate generator. The concurrent CODA [11] also selects, over annotated system and bar regions with a cascade trained end to end, and contributes a discontinuity benchmark we adopt in Sec. 3.3; we select over note hypotheses the generator already emits, so nothing is retrained and no layout annotation is read. Set-based scoring follows [16], and re-scoring detections has precedent in vision [17, 18]. Conditioning on its own past choices exposes the selector to the drift of sequence prediction [19, 20], which we quantify.

2. SCORE FOLLOWING AS SEQUENTIAL SELECTION

2.1. Hypotheses

A performance is processed in instants $t = 1, \dots, T$. At each instant a candidate generator reads the audio so far and the current page and emits note hypotheses $b_i = (x_i, y_i, w_i, h_i)$ with confidence o_i ; we keep the $K = 256$ most confident, $\mathcal{C}_t$. Staves are concatenated along one horizontal axis, the unrolled coordinate u_i , so that positions on either side of a line break are adjacent. Any note detector can serve; we use released ones [9, 10], which emit 8,112 hypotheses per instant on a 52×52 grid, and never update their weights. Reporting the most confident hypothesis,

$$\hat{y}_t = \arg \max_{i \in \mathcal{C}_t} o_i, \quad (1)$$

is the degenerate selector, and is what a detection-based regressor does. CANDOR reads $\mathcal{C}_t$ and its own previous choice $\hat{u}_{t-1} = u_{i_{t-1}}$, and nothing else.

2.2. Temporal constraint

Selections must form a plausible trajectory. Between instants Δ_t frames apart the performer advances by a roughly predictable amount, so we score the displacement $d_i = u_i - \hat{u}_{t-1}$ with a floored Gaussian whose mean grows with elapsed time,

$$\log \pi_i = \max \left(-\frac{(d_i - \mu \rho_t)^2}{2\sigma^2}, J \right), \quad (2)$$

with $\rho_t = \text{clip}(\Delta_t / \Delta_0, 0.2, 8)$ and $\Delta_0 = 5$ frames, the mean inter-onset gap in training. The floor J leaves jumps possible at bounded cost. None of (μ, σ, J) is learned. Combined with confidence, $h_i = \log o_i + \log \pi_i$, this alone is a competent selector (Table 2).

2.3. Set scorer

The prior treats hypotheses independently given the history. What it cannot express is comparative: whether *this* box, among the ones on offer right now, looks like a real note-head. Each hypothesis is therefore described by $\phi_i \in \mathbb{R}^{33}$ holding confidence relative to the instant's best, box geometry, motion relative to the history, relation to any coarse regions the generator emits, and a neighbourhood group (how many hypotheses lie within 25 and 100 px, spacing to the nearest on each side, rank in reading order, size against the instant's median). We also read the 128-dimensional feature f_i from which the generator scored the box. A permutation-invariant network [16] embeds the set and pools a context every hypothesis sees:

$$e_i = g([\phi_i; \text{ReLU}(W f_i)]), \quad c = \left[\frac{1}{K} \sum_j e_j; \max_j e_j \right], \quad (3)$$

$$s_i = r([e_i; c]), \quad (4)$$

with g and r two-layer perceptrons (64, 32 and 64, 1 units) and $W \in \mathbb{R}^{64 \times 128}$. The maximum is per dimension, so each hypothesis is judged against its competitors. The scorer has 23,203 parameters.

2.4. Training and decoding

Selection is supervised listwise. With ε_i the error in frames of hypothesis i , the target is $q_i \propto \exp(-\varepsilon_i/\tau)$, $\tau = 3$, and the loss is

$$\mathcal{L} = - \sum_{i \in \mathcal{C}_t} q_i \log \frac{\exp s_i}{\sum_{j \in \mathcal{C}_t} \exp s_j}, \quad (5)$$

so the scorer is trained to rank a set rather than to regress a coordinate. Training runs on the MSMD training pieces [21], synthesized and convolved with measured room impulse responses [22]. History features come from the previous instants' selections, perturbed on 30% of samples by Laplace noise of

scale 30 px. We use AdamW ($3 \cdot 10^{-3}$, cosine, 40 epochs, batch 256). At each instant CANDOR selects

$$\hat{i}_t = \arg \max_{i \in \mathcal{C}_t} \beta s_i + (1 - \beta) h_i, \quad (6)$$

and $\hat{i}_t$ becomes the history of instant $t + 1$. Decoding is causal and greedy: one forward pass over 256 hypotheses, 1.25 ms on a single CPU thread. Training the scorer on the residual $s_i - h_i$ and selecting $\arg \max_i (s_i + h_i)$ is Eq. 6 at $\beta = 0.5$ and removes all four scalars from the system; that variant is reported below.

3. EXPERIMENTS

Data and metric. Selectors are trained on 353 MSMD pieces [21] and validated on 19. The evaluation set is 16 pieces, 25 pages and 4,149 onsets played on a Yamaha Avant-Grand N2 and captured with a room microphone [7]. A held-out set of 80 further pieces is rendered with a real impulse response and additive noise from 12 to 0.5 dB SNR (40,594 onsets per level). We report the ratio of onsets whose selected position maps to within 0.05 to 5 s of the true time, 0.5 s being the primary figure, with 95% intervals bootstrapped over pieces [23].

Selection of the four scalars. (μ, σ, J, β) are fitted leave-one-piece-out: for each evaluation piece they are chosen on the other 15 with pieces weighted equally, and that piece is decoded with them, so no piece influences its own decoding. Every CANDOR number below is cross-validated this way, and the residual variant needs no such fit at all. Seeds move results by about a point, so rows are seed means with standard deviations.

3.1. Selection is separable from proposal

Table 1 reports CANDOR on real recordings and Table 2 decomposes it. The same formulation, with the selector retrained on whichever hypotheses the generator emits, holds across three generators of very different quality. It is worth 20.3 points on the weakest, whose training data is entirely public, lifting it past where the strongest generator begins, and reaches 94.4% on the strongest. A selector fitted to one kind of proposal does not need the proposals to be good; it needs them to contain the answer, which the oracle says they do.

Two rows keep the claim honest. The parameter-free prior alone accounts for roughly half the gain, so the learned component is not carrying the result by itself, and the residual variant, which fits no scalar to any evaluation data, reaches 88.7%. Selection is also cheap: 23k parameters and 1.25 ms per instant against 12.8 ms on a GPU for the closest comparable system [11].

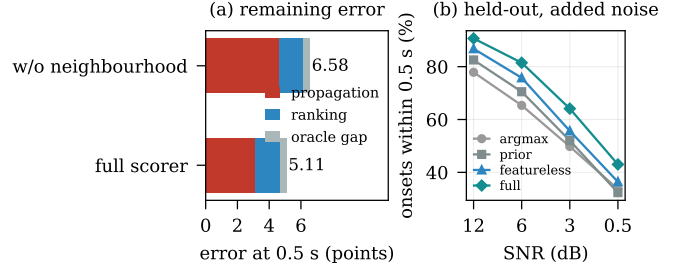


Fig. 2. (a) Error at 0.5 s split by teacher forcing into drift, ranking, and the gap to a causal oracle over the same hypotheses. (b) Held-out pieces with a real impulse response and additive noise, 40,594 onsets per level.

3.2. What selection needs, and what it does not

Not pixels. Feeding the selector the generator’s backbone features, DINOv2 patch features [24] at the same grid, or no image features at all gives 90.9 in all three cases. On synthesized validation audio the same three differ by 3.4 points, so the features are learnable and useful there and worth nothing where it counts. Replacing the generator’s audio encoder is no better: a MERT tower [25] costs more than 40 points, and **Mamba pending**. Perception is not the binding constraint, which is what the oracle predicted.

But history. Removing the features computed against the previous selection costs 7.2 points, the largest single ablation here. Teacher forcing attributes about two thirds of the remaining error to drift: supplied the true previous position the same selector reaches 98.0% against the causal oracle’s 99.6% (Fig. 2a). Selection is sequential, and it is the sequence that is hard.

Synthetic accuracy does not rank selectors. Across 48 trained checkpoints, accuracy on synthesized validation audio correlates $r = 0.16$ with accuracy on real recordings; the best-validation checkpoint scores 89.3 on real audio, below a selector reading no image features at all. A reverberant held-out set does better ($r = 0.80$) and still mispicks. This is why the four scalars are fitted on the target recordings under cross-validation rather than on a synthetic proxy, and it is a caution for the field.

3.3. Where the temporal constraint hurts

Eq. 2 assumes forward motion, which is why selection works and also why a repeat or a jump breaks it. We splice three discontinuities into each performance at random destinations, following the protocol of [11], re-running the generator on the spliced audio so its conditioning is genuinely wrong afterwards, and excluding jumps across a page boundary where our state resets anyway, leaving 41.

Table 3 reports a failure. CANDOR is worse after a jump than selecting by confidence alone, 0.15 against 0.22 of onsets tracked over the next five seconds, because the prior pins

Table 1. Real piano recordings, 16 pieces and 4,149 onsets captured with a room microphone. Ratio of onsets tracked within each error threshold. [†] generator trained with additional proprietary recordings, public checkpoint. *Layout*: needs system and bar annotations at test time. *Trains*: what the method fits on the target task. Published figures are as reported in [11].

		Onset error threshold (s)					Layout	Trains
		≤0.05	≤0.1	≤0.5	≤1.0	≤5.0		
ours	CANDOR, hypotheses from CYOLO	.	.	.916	.	.	no	23k selector
	CANDOR, hypotheses from CYOLO-SB	.	.	.909	.	.	no	23k selector
	CANDOR, hypotheses from CYOLO-SB+A [†]	.	.	.944	.	.	no	23k selector
prior work	CYOLO [9]	.563	.581	.712	.749	.919	no	1.5M detector
	CYOLO-SB [10]	.610	.630	.799	.832	.960	no	1.5M detector
	CYOLO-SB+A [†] [10]	.682	.706	.865	.891	.981	no	1.5M detector
	CODA [11]	.721	.743	.883	.914	.987	yes	2.0M end to end

Table 2. Components of CANDOR, all cross-validated per piece on the real recordings at 0.5 s. *argmax* is Eq. 1 over the same hypotheses, *prior* adds Eq. 2, *full* adds the scorer.

	<i>n</i>	<i>argmax</i>	<i>prior</i>	<i>full</i>
<i>Candidate generator</i>				
CYOLO [9]	3	71.3	72.5	91.6 ± 1.4
CYOLO-SB [10]	6	80.0	86.5	90.9 ± 0.6
CYOLO-SB+A [†]	3	86.5	90.8	94.4 ± 0.9
<i>Selector, on CYOLO-SB hypotheses</i>				
residual, no fitted scalars	3	80.0	86.5	88.7 ± 0.9
without temporal history	3	80.0	86.5	83.7 ± 0.4
<i>Image representation the selector reads</i>				
generator backbone (default)	6	80.0	86.5	90.9 ± 0.6
DINOv2 [24]	1	80.0	86.5	90.9
none	1	80.0	86.5	90.9
<i>Audio encoder inside the generator, retrained</i>				
MERT [25]	3	33.4	31.9	46.2 ± 9.1
Mamba [26]	3	.	.	.

the trajectory to a stale position. Suppressing the prior on detected silence recovers part of it when a break exists and nothing when it does not; triggering instead on the selector’s own confidence, which needs no break, works in both cases but only returns to the confidence baseline. A prior strong enough to be worth eleven points on continuous playing is strong enough to be wrong at a discontinuity, and handling both within one selector is open.

3.4. Robustness and budget

On held-out pieces (Fig. 2b) the scorer improves on the prior at every noise level, by less as the SNR falls: comparative evidence helps most while the hypotheses are still trustworthy. On 94 synthetic test pieces, where confidence alone selects .893 at 0.5 s, CANDOR reaches .955. Halving the hypothesis budget to $K = 128$ costs 0.5 points, and the correct hypothesis

Table 3. Recovery from 41 spliced discontinuities. *rec@1*: fraction first tracked correctly within 1 s. *post*: onsets tracked within 1 s over the following 5 s.

Selector	break present		no break	
	rec@1	post	rec@1	post
confidence only (1)	0.32	0.22	0.27	0.22
CANDOR	0.12	0.15	0.10	0.15
+ silence trigger	0.27	0.19	0.10	0.15
+ confidence trigger	0.15	0.22	0.20	0.23

sits at rank 27 at the 90th percentile, so the tail of the set is load-bearing.

4. CONCLUSION

Posing sheet-image score following as sequential selection over note hypotheses makes the synthetic-to-real gap measurable and mostly closable. The right answer is already in the set on real audio; a 23k-parameter selector with a parameter-free temporal prior finds it 94.4% of the time at 1.25 ms per instant, and transfers across candidate generators because selection and proposal are separate concerns. What the selector needs is its own history, not better pixels, and the temporal constraint it depends on is exactly what fails at a discontinuity. As in prior work the correct page is given, all performances are piano, and with 16 evaluation pieces differences of a point or two are not resolvable. Agreement between the pitches a hypothesis implies and those heard is a natural next source of evidence that does not depend on the history. Code and trained selectors will be released.

5. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical study that uses only publicly available data; no ethical approval was required.

6. ACKNOWLEDGMENTS

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